

Hypothesis Test & Statistical Analysis

T-test (one sample t-test)

1) In the population the average IQ is 100. A team of researchers want to test a new medication to see if it has either a positive or negative affect on intelligence, or no effect at all. A sample of 30 participants who have taken the medication has a mean of 140 with a std deviation of 20. Did the medication affect intelligence? $CI = 95\%$.

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$$\mu = 100, n = 30, \bar{x} = 140, CI = 95\%$$

$$\alpha = 1 - 0.95 = 0.05$$

1) Null Hypothesis $H_0 \Rightarrow \mu = 100$

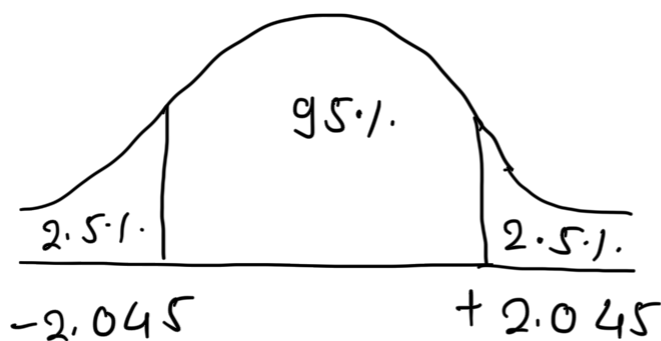
Alternate Hypothesis $H_1 \Rightarrow \mu \neq 100$

So this is 2-tail test

2) Significance value $\alpha = 0.05$

3) Degree of Freedom = $n - 1 = 30 - 1 = 29$
(How many choices
you have)

4) Decision Boundary



2.045 comes from t-table

Look For degree of Freedom here 29
on first column & α here 0.05 on
first row

If t -test is less than -2.045 or greater than 2.045 . Reject the null hypothesis

5) Calculate the t -test statistics

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{140 - 100}{20/\sqrt{30}} = \frac{40}{3.65} = 10.96$$

6) Conclusion

Decision Rule: IF t test is less than -2.0452 or greater than 2.0452 , reject the null

$t = 10.96 > 2.0452 \Rightarrow$ Reject the null Hypothesis